FRME-GNN | Research Progress and Experimental Evidence

Benchmark reproduction • Stage 1–4 implementation • Failed mask repairs • Alternative architectures • Revised proposal

Research status: 27 October 2026 | Presentation reference: 29 October 2026

Executive summary

This report consolidates completed work and the latest proposal. Five-seed benchmarking across Bail, Credit and German documented accuracy–fairness trade-offs. A masked GCN (A0–A5) was then implemented with fidelity, fairness, robustness and distillation objectives. Despite some strong classification results, edge masks frequently collapsed and matched random-mask replacements performed similarly. A repaired top-k mask and 90 runs across three alternative mask-training strategies did not establish reliable edge importance. The immediate research response is to validate graph signal and explanation fidelity, then evaluate a literature-grounded FairGNN + PGExplainer + GeoXGNN-inspired framework. That final integrated method is proposed, not yet empirically validated.

1. Problem statement and experimental goals

Original FRME-GNN goal: develop a graph neural network that combines competitive node classification, group fairness, faithful edge-level explanations and stability under graph perturbations. Crucially, an explanation is only useful if its selected edges contribute task-relevant evidence beyond chance or simple topology heuristics.

Refined goal: retain a validated GNN/explainer backbone and make targeted, independently testable improvements to (i) utility-preserving fairness, (ii) edge-ranking fidelity, and (iii) explanation robustness under prediction-preserving graph perturbations. AngleFair-GNN remains a separate, exploratory geometric research track.

2. Benchmarks reproduced: datasets, methods and findings

Datasets: Bail, Credit and German; seeds 0–4. Reproduced/checkpointed methods: NIFTY Vanilla, NIFTY SSF, FairGNN under two run settings, and EDITS. Results below are five-seed means. Lower demographic-parity (DP) and equality-of-opportunity (EO) gaps are preferred; F1 and AUC are higher-is-better.

Data	Baseline	F1	AUC	DP gap	EO gap
Bail	FairGNN (NIFTY run set)	0.547	0.925	0.001	0.001
Bail	NIFTY Vanilla	0.737	0.833	0.074	0.059
Bail	EDITS	0.785	0.862	0.103	0.089
Credit	NIFTY SSF	0.817	0.718	0.116	0.093
Credit	FairGNN (NIFTY run set)	0.738	0.735	0.162	0.151
Credit	EDITS	0.853	0.685	0.052	0.027
German	NIFTY SSF	0.808	0.679	0.156	0.116
German	FairGNN (NIFTY run set)	0.428	0.541	0.016	0.004

Interpretation: on Bail, FairGNN had near-zero disparity with F1 ≈ 0.547, while EDITS reached F1 ≈ 0.785 at a higher parity gap ≈ 0.103. On Credit, EDITS reached F1 ≈ 0.853 but AUC ≈ 0.685. German exhibited marked instability in some masked/FairGNN settings. Small gaps alone do not establish useful or nondegenerate classification; methods and protocols must be aligned before claiming comparative superiority.

3. Completed FRME-GNN model stages

Stage	Models	Purpose

1	A0 (GCN), A1 (learned mask), A2 (fidelity + mask regularization)	Does edge weighting preserve predictions and select meaningful connections?
2	A3 (+ prediction / mask fairness penalties)	Can group disparity fall without a substantial utility cost?
3	A4 (+ perturbation consistency)	Are predictions and explanations stable under graph edits?
4	A5 (+ teacher prediction distillation)	Does distillation improve performance or efficiency?

Earlier Stage 1–2 comparisons on a previous protocol had Bail A0/A1/A2/A3 F1 ≈ 0.695/0.785/0.710/0.712 and Credit ≈ 0.815/0.827/0.810/0.786. These values should not be merged with later v2 results because run protocols and saved outputs differed.

4. Stage 1–4 v2 results: prediction improves, explanations do not

Data	Model	F1	AUC	DP	EO	Degenerate seeds
Bail	A0	0.880	0.936	0.065	0.034	0/5
Bail	A1	0.902	0.952	0.064	0.029	5/5
Bail	A2	0.928	0.964	0.068	0.032	5/5
Bail	A3	0.881	0.930	0.070	0.043	5/5
Bail	A4	0.876	0.933	0.068	0.040	5/5
Bail	A5	0.878	0.935	0.068	0.040	5/5
Credit	A0	0.878	0.679	0.029	0.019	0/5
Credit	A1	0.878	0.674	0.035	0.020	4/5
Credit	A2	0.879	0.682	0.028	0.019	4/5
Credit	A3	0.877	0.680	0.024	0.015	4/5
Credit	A4	0.878	0.672	0.016	0.008	5/5
Credit	A5	0.878	0.689	0.024	0.016	4/5

Finding: Bail A2 achieved F1 ≈ 0.928 and AUC ≈ 0.964, but its random-mask replacement achieved essentially identical F1 (≈ 0.928). Bail A1–A5 masks were flagged degenerate in 5/5 seeds; Credit A4 was 5/5 and several other models 4/5. Fairness additions generally did not deliver a reliable Pareto improvement; robustness drift scores close to zero also occurred for unmasked/degenerate models. A5 lacked a consistent cross-dataset benefit. The classifier outputs remain informative, but a successful learned explanation mechanism was not demonstrated.

5. Repair attempts and diagnostic experimental controls

Modifications implemented/tested: small nonzero mask-score initialization; warm-up; mean-weight retention penalties and then exact top-k retention; explicit mask-gradient and saturation logging; matched-budget random/degree controls; an MLP comparator and degree-preserving rewired-graph GCN; A2 fidelity/budget ablations; and broader metrics (AUROC, balanced accuracy, MCC, AUPRC and confusion matrices). Exact top-k prevents selecting every edge, but cannot force the scorer to learn meaningful ranks.

6. Latest 90-run Option A evaluation

Archive: diff_arch_results.zip. Three mask variants × three budgets (30%, 50%, 70%) × five seeds × two datasets = 90 runs. Results below are means at 50% retention; "Δ random" is learned-mask AUC minus a random-mask control using the same trained model.

Method	Bail AUC	Bail Δ random	Credit AUC	Credit Δ random
Independent edge scoring	0.9708	+0.0350	0.7142	−0.0042
Alternating bilevel (first-order)	0.9686	+0.0368	0.7109	−0.0039
Perturbation-based learning	0.9693	+0.0383	0.7091	−0.0044

Crucial failure: score standard deviation was < 0.0001 in 45/45 Bail runs and 33/45 Credit runs. Thus Bail top-k gains may arise from almost-tied scores and deterministic tie-breaking; they do not establish task-relevant ranking. Credit showed negative mean learned-minus-random AUC across the reported comparisons. These methods are implemented explorations, not three new message-passing backbones.

7. Graph-signal test: what actually succeeded

Stage 0 comparison	Bail AUC / F1	Credit AUC / F1
Always-positive	0.5000 / 0.5469	0.5000 / 0.8756
MLP (no graph)	0.9919 / 0.9786	0.6466 / 0.8664
GCN (original graph)	0.9311 / 0.8463	0.7070 / 0.8765
GCN (rewired graph)	0.6575 / 0.1204	0.5728 / 0.8744

Bail: feature-only MLP strongly outperformed the tested GCN, and removing graph messages from one trained masked model raised its AUC to ≈ 0.9902. Audit the feature pipeline and possible label leakage before interpreting structural explanations. Credit: original GCN AUC ≈ 0.707 exceeded MLP ≈ 0.647 and rewired GCN ≈ 0.573, giving preliminary evidence of useful topology. Credit always-positive F1 ≈ 0.876 shows why high F1 alone cannot validate the classifier. Rewiring-induced training collapse remains a possible confound.

8. Why the objectives were not achieved

- Mask optimization did not reliably separate edge scores; sigmoid saturation and near-ties persisted despite revised training strategies.
- Exact-budget top-k fixed the number of selected edges but did not establish mask importance, fidelity, or edge-order invariance.
- A2 classification gains were not causally attributable to learned masking; teacher fidelity, regularization, optimization and scaling remain possible explanations.
- Fairness gaps sometimes fell without stronger prediction or mask evidence; Stage 3 perturbation stability was not demonstrated beyond trivial baselines.
- The latest repaired Stage 1 CSVs reportedly contain inconsistent row widths (27 versus 29 fields); repair the schema and re-audit aggregated comparisons.

9. Revised proposed literature-grounded FRME-GNN method

Presentation-stage proposal only — not yet implemented or evaluated as a complete integrated system.

Published foundation	Known methodological role	Targeted FRME improvement to test
FairGNN (WSDM 2021)	Prediction-level graph fairness	Fairness-tolerance / utility-preservation comparison against fixed penalties
PGExplainer (NeurIPS 2020)	Parameterized graph explanation masks	Explicit rank/fidelity diagnostics; matched random/degree controls and planted-edge validation
GeoXGNN (IJCAI 2026, to verify bibliographic details)	Geometry-aware robust explanation motivation	Prediction-preserving graph-edit stability tested alongside explanation fidelity

Proposed stages: B0 = node classifier; B1 = parameterized explainer; B2 = B1 + fairness objective; B3 = B1 + robustness objective; B4 = B1 + fairness + robustness. Keep the same graph/splits, comparable model capacity, seeds, masks and optimization budgets. The method is a proposed integration/extension, not yet a validated new GNN backbone.

10. Next required experiments and decision gates

- 1. Reproduce baseline GCN twice with frozen splits; check preprocessing, feature leakage and logging.
- 2. On a planted-structure graph, show that learned edge scores recover known edges and are invariant to edge-list permutation; inspect top-k gradients and score variation.

- 3. On Credit, compare PGExplainer-inspired masks to random, degree and spectral-prior masks at equal edge counts using sufficiency/comprehensiveness, AUC/AUPRC/MCC and paired seeds.
- 4. Audit Bail as a potential feature-dominated negative control rather than forcing a structural explanation claim.
- 5. Only if mask integrity passes, evaluate FairGNN-inspired utility-preserving fairness and GeoXGNN-inspired stability as separate ablations, then their combination.
- 6. Run GATv1/GATv2, GIN and GraphSAGE only after the core mask mechanism is validated; currently GCN is the implemented backbone.

11. Final status for the presentation

Completed: five-seed benchmark checks; A0–A5 experiments; mask-collapse diagnosis; Stage 0 graph-signal comparisons; 90-run alternative-mask sweep; initial failure-aware research tooling. Not achieved: reliable informative edge masks, clear fairness–utility advantage, robust explanation gains or validated novel propagation operator. Proposed next: published-method-based FRME-GNN integrating FairGNN-inspired fairness, PGExplainer-inspired rank-faithful explanations and GeoXGNN-inspired robustness, with strict ablations. AngleFair-GNN is separate and its real-data analysis must wait for nondegenerate masks.

Evidence and reporting notes

Internal source files used: all_baselines_summary_mean_std(1).csv; all_stages_summary.csv; diff_arch_results.zip and FRME_GNN_Latest_Results_Evaluation_Report.pdf. Multiple experimental generations are kept separate because their setup and model outputs differ. Numbers are reported as archive-derived five-seed means; uncertainty and matched-seed tests should accompany any paper submission. Research claims regarding bibliographic novelty and GeoXGNN details require primary-source verification before public publication.